

INTERNSHIP WORK SUMMARY

Prajwal Gawande | DevOps Intern | 12th Wonder

Introduction

During my internship at 12th Wonder as a DevOps Engineer Intern, I worked on real-world DevOps projects involving Azure DevOps, TeamCity, Docker, Kubernetes, ArgoCD, Harbor Registry, SonarQube, Snyk, Grafana, Prometheus, Alertmanager, PostgreSQL, Git, Bitbucket, and Linux.

I contributed to multiple enterprise applications, including Titan, Titan 4511, Sustainium, TechConnector, Absorb LMS, PMI-SSO, Sefrium, and other Titan environments. My responsibilities included designing, developing, maintaining, and troubleshooting CI/CD pipelines, automating application deployments, implementing security scanning, configuring monitoring solutions, and supporting production deployments.

A significant part of my internship involved migrating the Titan deployment environment from TeamCity to Azure DevOps. I redesigned and automated the complete deployment workflow by creating Azure DevOps CI/CD pipelines that automatically cloned source code, restored dependencies, built the application, published artifacts, built Docker images, pushed images to Harbor Registry, updated Kubernetes deployment manifests, triggered ArgoCD synchronization, and deployed the latest application version to Kubernetes with minimal manual intervention.

I also focused on pipeline optimization by reducing pipeline execution time and improving deployment efficiency through build optimization, Docker image creation, artifact generation, and deployment automation.

I am deeply grateful to my manager, Mr. Rajesh Vishwakarma, whose guidance, training, and continuous support helped me build strong DevOps knowledge and gain hands-on experience with production-grade infrastructure.

Key Contributions

- Successfully migrated the Titan deployment environment from TeamCity to Azure DevOps.
- Designed, created, and maintained CI/CD pipelines for multiple enterprise applications and databases.
- Automated the complete deployment workflow from source code to production deployment.
- Configured pipelines to automatically clone source code, restore dependencies, build the project, publish artifacts, build Docker images, push images to Harbor Registry, update Kubernetes manifests, trigger ArgoCD synchronization, and deploy applications to Kubernetes.
- Optimized Azure DevOps pipeline execution time to achieve significantly faster builds and deployments.
- Built CI/CD pipelines for Titan, Sustainium, TechConnector, database deployments, and additional environments.
- Integrated SonarQube and Snyk security scanning into CI/CD pipelines.
- Configured Grafana, Prometheus, and Alertmanager with Microsoft Teams notifications.
- Troubleshot production issues related to Azure DevOps, Kubernetes, Docker, PostgreSQL, SonarQube, and deployment pipelines.

Core Technologies

- Azure DevOps
- TeamCity
- Docker
- Harbor Registry
- Kubernetes
- ArgoCD
- Git

- Bitbucket
- Linux
- PostgreSQL
- SonarQube
- Snyk
- Grafana
- Prometheus
- Alertmanager

Skills Demonstrated

Technical Skills

- CI/CD pipeline design, development, and maintenance (Azure DevOps)
- Legacy infrastructure migration (TeamCity → Azure DevOps)
- Docker containerization and image management via Harbor Registry
- Kubernetes deployment management and manifest updates
- GitOps-based deployment automation using ArgoCD
- Security scanning integration (SonarQube, Snyk) within CI/CD pipelines
- Monitoring and alerting setup (Grafana, Prometheus, Alertmanager) with Microsoft Teams integration
- Database deployment and management (PostgreSQL)
- Secrets management using Azure Key Vault
- Version control and source code management (Git, Bitbucket)
- Linux system administration

Professional Skills

- End-to-end ownership of deployment automation across multiple enterprise applications
- Root cause analysis and troubleshooting of production issues
- Working across multiple project environments and applications simultaneously
- Performance-oriented mindset — continuously optimizing for speed and efficiency
- Cross-functional coordination while supporting production deployments

Key Achievements

- Led the end-to-end migration of the Titan deployment environment from TeamCity to Azure DevOps, redesigning the deployment workflow from source code to production.
- Built a fully automated CI/CD pipeline covering source cloning, dependency restore, build, artifact publishing, Docker image build, Harbor registry push, Kubernetes manifest updates, and ArgoCD sync — requiring minimal manual intervention.
- Significantly reduced pipeline execution time through build, Docker image, and deployment optimization.
- Delivered CI/CD pipelines across 8+ enterprise applications, including Titan, Titan 4511, Sustainium, TechConnector, Absorb LMS, PMI-SSO, and Sefrium.
- Strengthened security posture by integrating SonarQube and Snyk scanning directly into the pipeline, enabling early detection of code quality and vulnerability issues.
- Set up end-to-end monitoring and alerting with Grafana, Prometheus, and Alertmanager, with real-time notifications routed to Microsoft Teams.

- Resolved production issues across the stack — Azure DevOps, Kubernetes, Docker, PostgreSQL, and SonarQube — ensuring deployment stability.
- Achieved complete end-to-end pipeline automation — covering project build, Docker image creation, Harbor registry push, Kubernetes manifest update, and ArgoCD sync — with the entire deployment cycle completing in under 5 minutes with zero manual intervention.

Acknowledgement

I would like to express my sincere gratitude to Mr. Rajesh Vishwakarma, my manager at 12th Wonder, for his exceptional guidance, training, and continuous support throughout this internship. His mentorship helped me build strong DevOps fundamentals and gain valuable hands-on experience with production-grade infrastructure, and I am truly grateful for the opportunity to learn under his guidance.

Conclusion

Throughout this internship, I have gained comprehensive, hands-on experience across the full DevOps lifecycle — from CI/CD pipeline design and containerization to Kubernetes orchestration, GitOps automation, security scanning, and production monitoring. Leading the Titan migration from TeamCity to Azure DevOps gave me the opportunity to take ownership of a real, high-impact infrastructure project from design through to production deployment.

I am motivated to continue contributing to 12th Wonder's DevOps practices, applying the skills and production experience I've built during this internship, and I would welcome the opportunity to transition into a full-time role with the team.

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